

Linear mixed models in R

Day 4

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Error messages during model reduction

When reducing a maximal model (or running models in general) you can encounter the following error message:

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl =  
control$checkConv, :  
## Model failed to converge with max|grad| = 0.609847 (tol = 0.002,  
component 1)  
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl =  
control$checkConv, : Model is nearly unidentifiable: very large  
eigenvalue  
## - Rescale variables?
```

This might indicate a problem with your data set, usually a variable that is not normally distribut.

Common transformations

Often independent variables are also transformed for use in LMEs

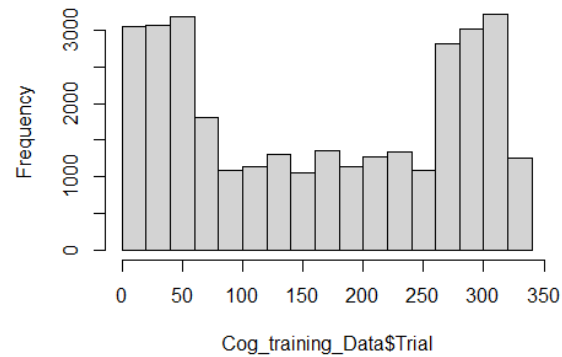
Common transformations for independent variables are:

- Logarithmic
- Reciprocal
- Scaling

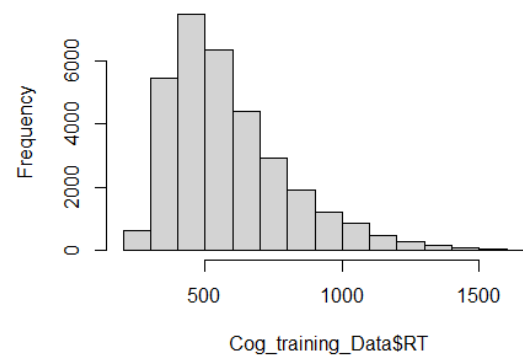
BUT much more important is the normal distribution of residuals (see Day 2)!

Common transformations

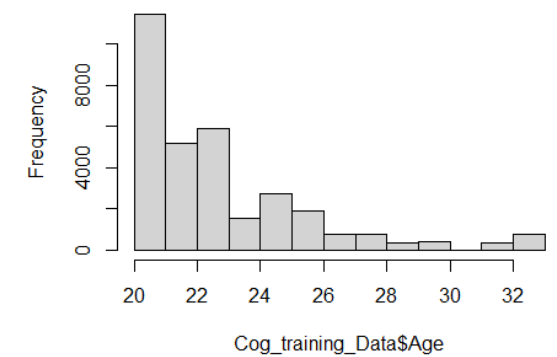
Histogram of Cog_training_Data\$Trial



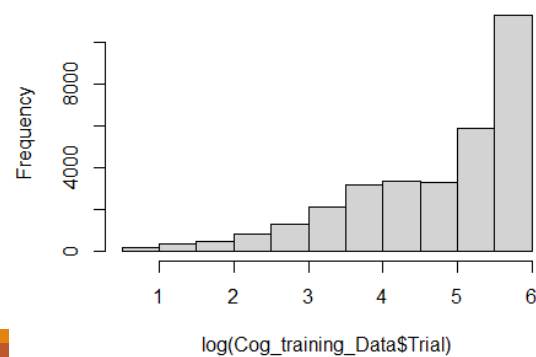
Histogram of Cog_training_Data\$RT



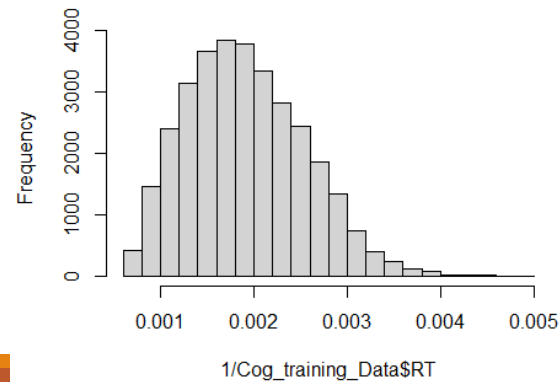
Histogram of Cog_training_Data\$Age



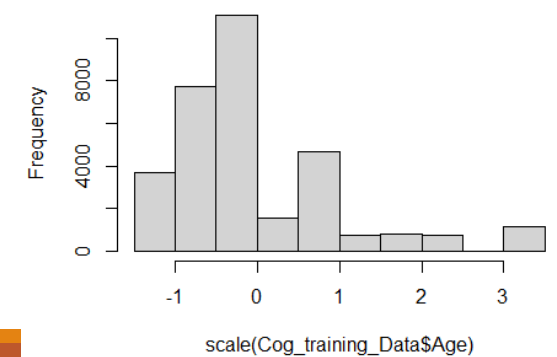
Histogram of log(Cog_training_Data\$Trial)



Histogram of 1/Cog_training_Data\$RT



Histogram of scale(Cog_training_Data\$Age)



Shapiro-Wilk Test

Statistical test for checking if a variable is normally distributed

The sample size must be between 3 and 5,000

P-value below 0.05 indicates non-normally distributed data

```
shapiro.test(unique(PN_Data3$lg.freq))
```

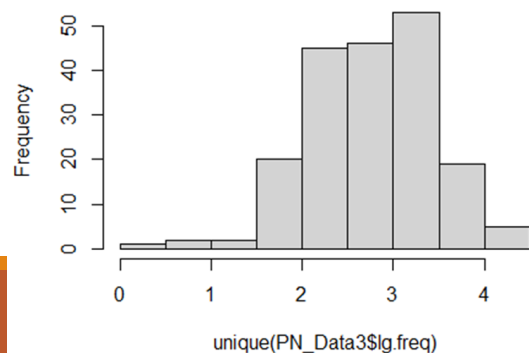
```
##
```

```
## Shapiro-Wilk normality test
```

```
## data: unique(PN_Data3$lg.freq)
```

```
## W = 0.98921, p-value = 0.1534
```

Histogram of unique(PN_Data3\$lg.freq)



```
shapiro.test(unique(PN_Data3$Trial))
```

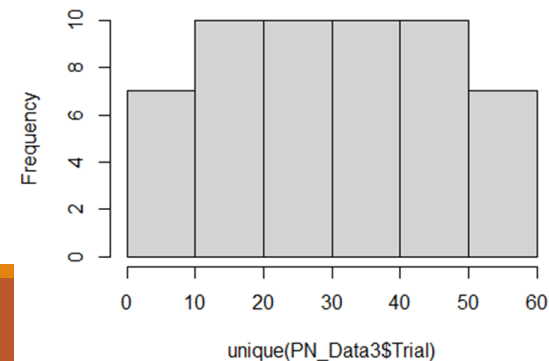
```
##
```

```
## Shapiro-Wilk normality test
```

```
## data: unique(PN_Data3$Trial)
```

```
## W = 0.95542, p-value = 0.04305
```

Histogram of unique(PN_Data3\$Trial)



Lo and Andrews (2015) To transform or not to transform: using generalized linear mixed models to analyse reaction time data

Transforming your data can obscure or create effects in your model results

Makes interpretation much more difficult/changes it

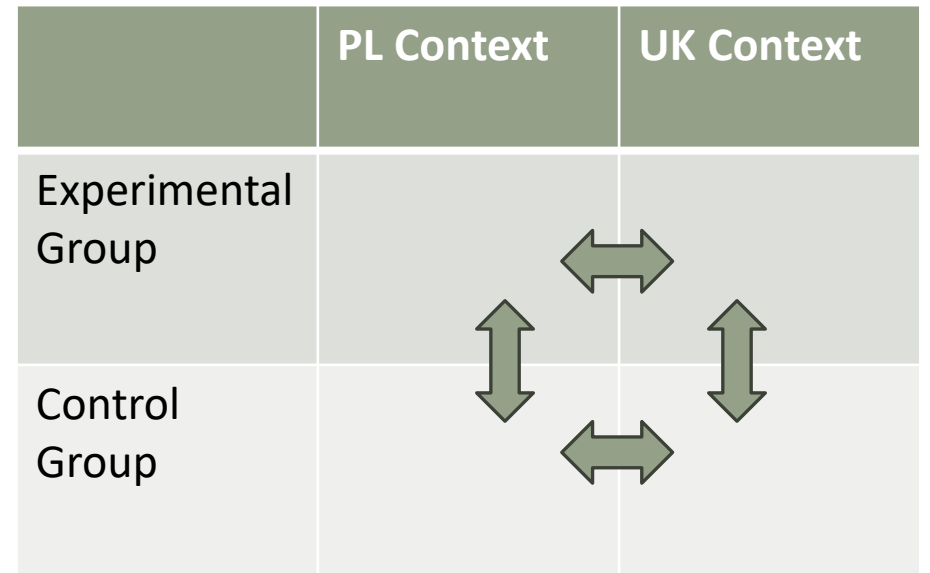
GLMs might be a better tool to account for non-normal distributed data

Understanding model output

```
summary(model_Large3)
```

```
## Fixed effects:
```

##	Estimate	Std. Error	t value
## (Intercept)	1020.739	23.344	43.726
## GroupExperimental	-76.150	29.565	-2.576
## ContextUK	-18.660	8.471	-2.203
## GroupExperimental:ContextUK	53.089	11.706	4.535



Emmeans-package

```
library(emmeans)

em1 <- emmeans(model_Large3, specs = pairwise ~ Group:Context)

em1$emmeans

##   Group      Context emmean    SE  df asymp.LCL asymp.UCL
##   Control      PL      1021 23.3 Inf        975      1066
##   Experimental PL      945 23.5 Inf        899        991
##   Control      UK      1002 23.3 Inf        956      1048
##   Experimental UK      979 23.4 Inf        933      1025
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

Computes estimated marginal means (EMMs) for specified factors or factor combinations in a linear model and comparisons or contrasts among them

EMMs are also known as least-squares means (minimized sum of squared residuals)

Emmeans-package

```
em1$contrasts
```

##	contrast	estimate	SE	df	z.ratio	p.value
##	Control PL - Experimental PL	76.1	29.57	Inf	2.576	0.0491
##	Control PL - Control UK	18.7	8.47	Inf	2.203	0.1224
##	Control PL - Experimental UK	41.7	29.55	Inf	1.412	0.4919
##	Experimental PL - Control UK	-57.5	29.51	Inf	-1.948	0.2080
##	Experimental PL - Experimental UK	-34.4	8.04	Inf	-4.282	0.0001
##	Control UK - Experimental UK	23.1	29.51	Inf	0.782	0.8629
##						
##	Degrees-of-freedom method: asymptotic					
##	P value adjustment: tukey method for comparing a family of 4 estimates					

Understanding model output

Em1\$emmeans

##	Experimental PL	945	23.5	Inf	899	991
##	Experimental UK	979	23.4	Inf	933	1025

Em1\$contrasts

##	Experimental PL - Experimental UK	-34.4	8.04	Inf	-4.282	0.0001
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945ms – 979ms = -34ms

Why the weird variable names?

```
summary(model_Large3)
```

```
## Fixed effects:
```

##	Estimate	Std. Error	t value
## (Intercept)	1020.739	23.344	43.726
## GroupExperimental	-76.150	29.565	-2.576
## ContextUK	-18.660	8.471	-2.203
## GroupExperimental:ContextUK	53.089	11.706	4.535

Lme4 uses contrast-coding or dummy-coding for categorical variables

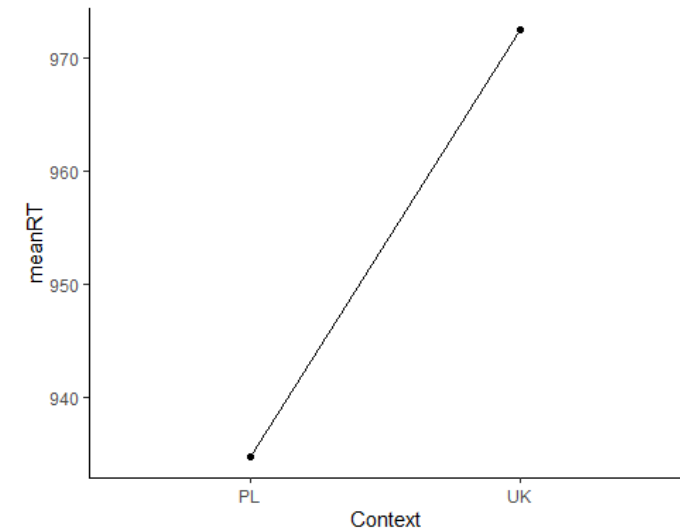
Contrasts have two functions:

- Control estimate calculation
- Handling multilevel categories

The explanatory variables are related linearly to the response.

Linear mixed effects models are **linear** and additive

Categorical variables are treated as 2-level variables



Contrasts

Categorical variables are used as factorial predictors, with several categorical levels

To estimate their effects, we assign numbers to each of the levels

These number are used for calculating comparisons

Different contrasts express different hypotheses

The underlying model remains the same, only the parametrisation of the effects changes

-> The choice of contrasts has no statistical consequence

-> It only changes estimate and hypothesis interpretation

Schad, D. J., Vasishth, S., Hohenstein, S., & Kliegl, R. (2020). How to capitalize on a priori contrasts in linear (mixed) models: A tutorial.

Implementing specific hypotheses into your model

Contrasts *reparametrize* the model and changes interpretation of parameters

R by default orders factors alphabetically and uses the first level as the baseline

- Contrasts are the re-ordering of levels

Wide variety of contrast types

- Treatment contrasts
- Sum contrasts
- Repeated contrasts
- Polynomial contrasts

Treatment contrasts

Default setting of R

- > Categorical variables are sorted alphabetically and turned into treatment contrasts
- > Category levels are 'dummy-coded' with 0/1 values
- > The level coded as 0 is the 'reference level' or 'baseline level'

Contrast implementation in R

```
contrasts(PN_Data$Context)
```

```
##      UK
```

```
## PL    0
```

```
## UK    1
```

```
PN_Data$Context <- factor(PN_Data$Context, levels = c("UK", "PL"))  
contrasts(PN_Data$Context)
```

```
##      PL
```

```
## UK    0
```

```
## PL    1
```


Treatment contrasts

```
##      UK
```

```
## PL   0
```

```
## UK   1
```

```
model_Large8 = lmer(RT ~ Group*Context + (1 | Subject) +  
                    (1 | ItemNr), data=PN_Data)
```

```
summary(model_Large8)
```

```
## Fixed effects:
```

##	Estimate	Std. Error	t value
## (Intercept)	1020.81	23.42	43.589
## GroupExperimental	-76.20	29.45	-2.587
## ContextUK	-18.74	8.49	-2.208
## GroupExperimental:ContextUK	53.00	11.73	4.517

```
##      PL
```

```
## UK   0
```

```
## PL   1
```

```
model_Large8 = lmer(RT ~ Group*Context + (1 | Subject) +  
                    (1 | ItemNr), data=PN_Data)
```

```
summary(model_Large8)
```

```
## Fixed effects:
```

##	Estimate	Std. Error	t value
## (Intercept)	1002.07	23.36	42.895
## GroupExperimental	-23.19	29.40	-0.789
## ContextPL	18.74	8.49	2.208
## GroupExperimental:ContextPL	-53.00	11.73	-4.517

Treatment contrasts

```
eml$contrasts
```

##	contrast	estimate	SE	df	z.ratio	p.value
##	Control PL - Experimental PL	76.1	29.57	Inf	2.576	0.0491
##	Control UK - Experimental UK	23.1	29.51	Inf	0.782	0.8629

```
eml$emmeans
```

##	Group	Context	emmean	SE	df	asympt.LCL	asympt.UCL
##	Control	PL	1021	23.3	Inf	975	1066
##	Control	UK	1002	23.3	Inf	956	1048

```
## Fixed effects:
```

##	Estimate	Std. Error	t value
## (Intercept)	1020.81	23.42	43.589
## GroupExperimental	-76.20	29.45	-2.587
## ContextUK	-18.74	8.49	-2.208
## GroupExperimental:ContextUK	53.00	11.73	4.517

```
## Fixed effects:
```

##	Estimate	Std. Error	t value
## (Intercept)	1002.07	23.36	42.895
## GroupExperimental	-23.19	29.40	-0.789
## ContextPL	18.74	8.49	2.208
## GroupExperimental:ContextPL	-53.00	11.73	-4.517

Using different contrasts

Intercepts estimate the dependent variable when *all* predictors are at 0

If the 'distance' between the levels equals 1, then the slope estimates their difference

->estimates measure the difference *per unit* of the predictor

Sum contrasts

If all predictors have their mean at 0, then the intercept estimates the grand-mean (across levels)

-> You are centering your predictor on the mean

-> Sum contrasts use $-0.5/0.5$; $-1/1$; $-2/2$ etc for centering

$-0.5/0.5$ also maintains distance between levels at 1

Sum contrasts

```
contrasts(PN_Data$Context) <- contr.sum(2)/2
contrasts(PN_Data$Context)
```

```
##      [,1]
```

```
## UK    0.5
```

```
## PL   -0.5
```

```
## Fixed effects:
```

##	Estimate	Std. Error	t value
## (Intercept)	1011.44	23.00	43.973
## GroupExperimental	-49.70	28.84	-1.723
## Context1	-18.74	8.49	-2.208
## GroupExperimental:Context1	53.00	11.73	4.517

```
##      UK
```

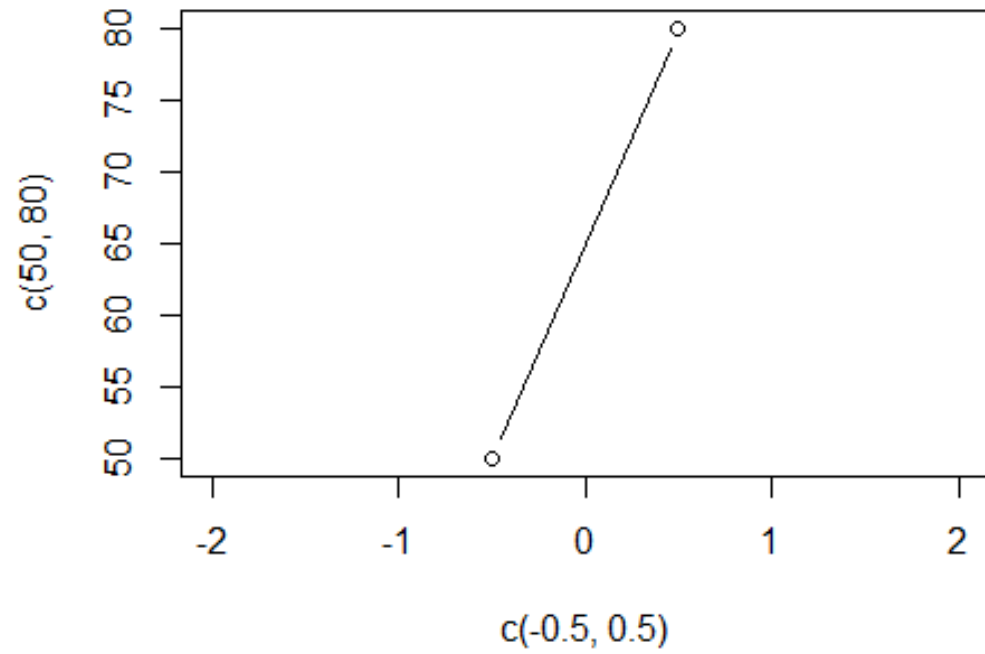
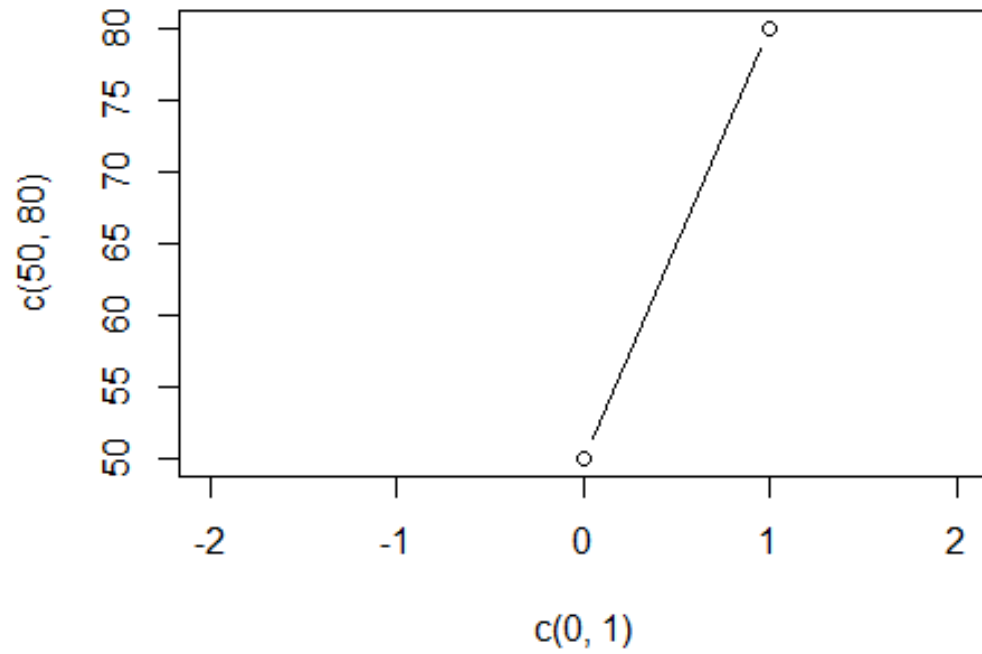
```
## PL    0
```

```
## UK    1
```

```
## Fixed effects:
```

##	Estimate	Std. Error	t value
## (Intercept)	1020.81	23.42	43.589
## GroupExperimental	-76.20	29.45	-2.587
## ContextUK	-18.74	8.49	-2.208
## GroupExperimental:ContextUK	53.00	11.73	4.517

Treatment and sum contrasts



Contrasts for multilevel categories

Three level categories are more difficult

-> Contrasts include the creation of dummy variables for comparing all levels of a category with each other

```
## Fixed effects:
```

##	Estimate	Std. Error	t value
## (Intercept)	1011.44	23.00	43.973
## Context.1	-9.74	7.34	-1.784
## Context.2	-2.54	8.75	-2.984
## Context.3	-11.67	8.63	-2.648

-> R provides functions to automatically create different contrasts for multiple levels

-> part of the MASS package

```
library(MASS)
```

Multiple levels with Treatment contrasts

```
contrasts(PN_Data$Context) <-  
contr.treatment(5)
```

Control group against experimental groups

```
contr.treatment(5)
```

Default setting in R

```
##      2  3  4  5
```

```
## 1  0  0  0  0
```

```
## 2  1  0  0  0
```

First level is treated as control group

```
## 3  0  1  0  0
```

```
## 4  0  0  1  0
```

```
## 5  0  0  0  1
```

Every other level is compared against it

Rows are group levels

Columns are tested comparisons

Multiple levels with Sum contrasts

```
contr.sum(5)
```

##	[, 1]	[, 2]	[, 3]	[, 4]
## 1	1	0	0	0
## 2	0	1	0	0
## 3	0	0	1	0
## 4	0	0	0	1
## 5	-1	-1	-1	-1

Rows are group levels

Columns are tested comparisons

Experimental groups against grand average

Level with 1 is always compared against all others

Level 5 is set to -1, as it is already implicitly compared to all others

Multiple levels with Sum contrasts

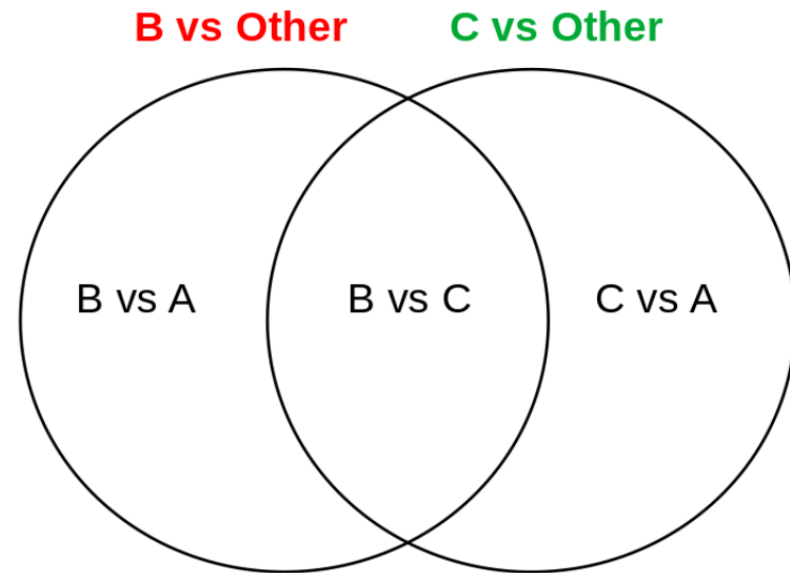
```
contr.sum(5)
```

##		[, 1]	[, 2]	[, 3]	[, 4]
##	1	1	0	0	0
##	2	0	1	0	0
##	3	0	0	1	0
##	4	0	0	0	1
##	5	-1	-1	-1	-1

Rows are group levels

Columns are tested comparisons

Three levels (A, B, C)



Multiple levels with repeated contrasts

```
contr.sdif(5)
```

##		2-1	3-2	4-3	5-4
##	1	-0.8	-0.6	-0.4	-0.2
##	2	0.2	-0.6	-0.4	-0.2
##	3	0.2	0.4	-0.4	-0.2
##	4	0.2	0.4	0.6	-0.2
##	5	0.2	0.4	0.6	0.8

Rows are group levels

Columns are tested comparisons

Comparisons are made between successive neighbouring levels

Requires ordered categories, but not evenly spaced ones

Centered around the grand average

Used for testing „increasing“ levels

Multiple levels with polynomial contrasts

```
contr.poly(5)
```

##		.L	.Q	.C	^4
##	[1,]	-6.324555e-01	0.5345225	-3.162278e-01	0.1195229
##	[2,]	-3.162278e-01	-0.2672612	6.324555e-01	-0.4780914
##	[3,]	-3.510833e-17	-0.5345225	1.755417e-16	0.7171372
##	[4,]	3.162278e-01	-0.2672612	-6.324555e-01	-0.4780914
##	[5,]	6.324555e-01	0.5345225	3.162278e-01	0.1195229

Checking for linear,
quadratic, cubical and
quartic trends in
categories

Required sorted and
evenly spaced categories

Rows are group levels

Columns are tested comparisons

Multiple levels with Helmert contrasts

```
contr.helmert(5)
```

Comparisons of the level with all previous levels

##		[, 1]	[, 2]	[, 3]	[, 4]
##	1	-1	-1	-1	-1
##	2	1	-1	-1	-1
##	3	0	2	-1	-1
##	4	0	0	3	-1
##	5	0	0	0	4

Rows are group levels

Columns are tested comparisons

LME to-do list

1 Hypotheses

Generate hypotheses based on previous research and your specific research questions

Collect data that might answer those questions

Formalize your hypotheses into falsifiable predictions

$$H_0 \text{ and } H_1$$

2 Prepare your data

In order to use data in your models, you need to prepare it correctly:

- Use the correct variable format (numeric, factor)
- Apply transformations if necessary (log, scaling, reciprocal)
- Apply contrasts for answering your research questions

```
load("PictureNaming.RData")
```

```
Model_Data <-  
  PN_Data %>%  
  mutate(Context = as.factor(Context)) %>%  
  select(Subject, Context, RT, Trial) %>%  
  mutate(Trial = as.numeric(Trial)) %>%  
  mutate(trans_RT = 1/RT)
```

```
contrasts(Model_Data$Context) <- contr.sum(2)/2
```


3 Build your model

Create the maximal model based on your data structure (clustering variables) and all available sensible data

Reduce the model according to Barr et al (2013)

Check your model assumptions

- Linearity
- Constant variance for residuals
- Normal distribution of residuals
- If necessary go back to step 2

4 Analyse your model

Use `summary()` to get your model output

- If desired with p-values using `lmerTest`

Understand and interpret your estimates of all fixed effects

Perform post-hoc test for (significant) interactions to understand the underlying driving effect

5 Report your results

Describe your data, your maximal and final model and your final statistical analyses

Create plots based on your model output

Questions and discussions

If you have anymore questions regarding previous lectures or your own research, please send me a mail to:

Jonas.walther@uni-tuebingen.de

We can discuss them during the course on Friday.

Evaluation

Please give your feedback for the course at:

<https://evaluation.zeq.uni-tuebingen.de/evasys/public/online/>



Thank you
for your
attention!
